

EXP.NO-7	Data Visualization Using Tableau

AIM:

To understand the Tableau interface and perform data visualization by connecting to a CSV data source. The experiment includes creating bar, line, and pie charts, adding a calculated field, and building an interactive dashboard and story.

PROCEDURE:

1. Launch Tableau & Explore Interface

- Open Tableau Desktop/Public.
- Explore the Start Page, Data Pane, Worksheet, Dashboard, and Story options.
- Identify the Columns shelf, Rows shelf, Marks card, Filters, and Show Me panel.

2. Connect to Data

- Click **Connect → Text File**.
- Select the Tableau_SalesData.csv file.
- Verify the fields such as Date, Region, Category, Sales, Profit, Quantity, and Target.

3. Create Basic Visualizations

- Create a **Bar Chart** by placing Category on Columns and Sales on Rows.
- Create a **Line Chart** by placing Date on Columns and Sales on Rows.
- Create a **Pie Chart** using Region and Sales.

4. Create a Calculated Field

- Select **Analysis → Create Calculated Field**.
- Create the calculated field **Profit Margin** using: $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}]) * 100$
- Create a bar chart showing Profit Margin by Category.

5. Build Interactive Dashboard

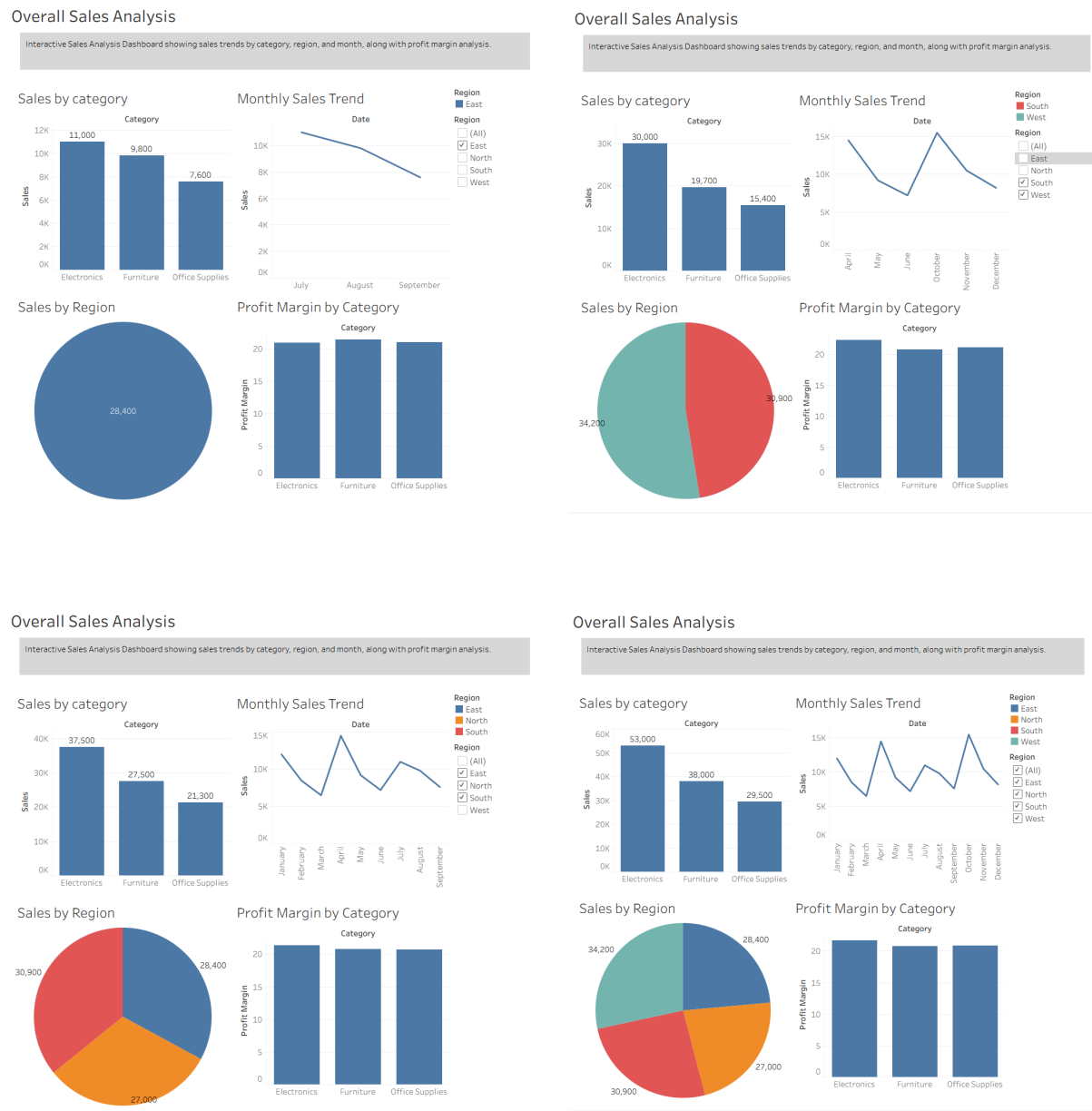
- Create a new dashboard.
- Add the Sales by Category, Monthly Sales Trend, Sales by Region, and Profit Margin by Category charts.
- Add a **Region filter** and apply it to worksheets using the same data source.

iv. Arrange and resize the charts to form a 2x2 dashboard layout.

6. Create Story

- Create a new Story.
- Add the **Sales Analytics Dashboard** to the story.
- Set the story title as **Overall Sales Analysis**.

OUTPUT:



RESULT:

Thus, the given data was successfully connected and visualized using Tableau. Bar, line, and pie charts were created along with a calculated field, interactive dashboard, and story successfully.